Dejan Gajic CV

Contact information

Name: Dejan Gajic

Place of birth: Sarajevo, Bosnia and Herzegovina

Nationality: Dutch

Address: Universität Leipzig
Fakultät für Physik und Erdsystemwissenschaften
Institut für Theoretische Physik
Brüderstrasse 16
04103 Leipzig
Germany

Email: dejan.gajic@uni-leipzig.de

URL: `https://home.uni-leipzig.de/gajic/`

Positions

W2 Professor in Mathematical Physics with tenure track to W3, Leipzig University, 2022–

Tenure track Assistant Professor in Mathematics, Radboud University, 2021–2022

Junior Research Fellow, University of Cambridge, King's College and DPMMS, 2017–2021

Chapman Fellow, Department of Mathematics, Imperial College London, 2016–2017

Education

Ph.D. in Mathematics,
University of Cambridge, 2012–2016,
Doctoral thesis adviser: Mihalis Dafermos

M.A.St. in Mathematics (Part III of the Mathematical Tripos), *Distinction*,
University of Cambridge, 2011–2012.

BSc in Mathematics,
Utrecht University, 2008–2011.

BSc in Physics and Astronomy,
Utrecht University, 2008–2011.

Research interests

Partial differential equations, general relativity, differential geometry

Grants and scholarships

ERC Starting Grant 2023, *The Mathematical Analysis of Extremal Black Holes and Gravitational Radiation* ("ExBHGravRad"), 1 497 500 EUR

Cambridge Home/EU Scholarships Scheme (CHESS), 2012–2015

Prins Bernhard Cultuurfonds, 2012–2015

Hendrik Muller Vaderlandschfonds, 2012–2015

Journal referee

Annals of Mathematics, Journal of the AMS, Communications in Mathematical Physics, Annals of PDE, Analysis & PDE, Annales Henri Poincaré, Publications Mathématiques de l'IHÉS, Annales scientifiques de l'École normale supérieure, Journal of Hyperbolic Partial Differential Equations, Astérisque, Duke Mathematical Journal, Probability and Mathematical Physics, Journal of Function Analysis

Teaching

Lecturer, *Black Holes*, Leipzig University, 2024–2025

Lecturer, *Group Theory and Its Applications in Physics*, Leipzig University, 2023

Lecturer, *Mathematical Physics*, Leipzig University, 2022–2025

Lecturer, *Nonlinear Wave Equations*, Radboud University, 2022

Lecturer, *Nonlinear Wave Equations in General Relativity*, University of Cambridge, 2019

Lecturer, *Functional Analysis*, Imperial College London, 2017

Teaching assistant, courses in analysis, geometry, topology, partial differential equations and black holes, University of Cambridge, 2012–2016, 2017–2020

Teaching assistant, courses in classical mechanics, electromagnetism, quantum mechanics and special relativity, Utrecht University, 2009–2010

Supervision

PhD:

Benedikt Miethke, Leipzig University, 2024–2027

Quinten Rutgers, Leipzig University, 2022–2025

MSc:

Inés Vega González, Leipzig University, 2024

Thomas Oppermann, Leipzig University, 2023–2024

Adam Almakroudi, Imperial College London, 2017

Seminar organization

General Relativity and PDE seminar, Leipzig University, 2022–

Mathematical Physics seminar, Radboud University, 2021–2022

General Relativity seminar, University of Cambridge, 2019–2020

Analysis seminar, Imperial College London, 2016–2017

Research publications

Published papers

1. Gajic, D. Late-time asymptotics for geometric wave equations with inverse-square potentials. *Journal of Functional Analysis 285*, 7 (2023), 110058.
2. Gajic, D., and Kehrberger, L. M. A. On the Relation Between Asymptotic Charges, the Failure of Peeling and Late-time Tails. *Classical and Quantum Gravity 39*, 19 (2022), 195006.
3. Angelopoulos, Y., Aretakis, S., and Gajic, D. Late-time tails and mode coupling of linear waves on Kerr spacetimes. *Advances in Mathematics 417* (2023), 108939.
4. Angelopoulos, Y., Aretakis, S., and Gajic, D. Price's law and precise late-time asymptotics for subextremal Reissner–Nordström black holes. *Annales Henri Poincaré 24V*, 9 (2023), 3215–3287.
5. An, X., and Gajic, D. Curvature blow-up rates in spherically symmetric gravitational collapse to a Schwarzschild black hole. *Archive for Rational Mechanics and Analysis* 247:51 (2023).
6. Gajic, D., and Warnick, C. Quasinormal modes in extremal Reissner–Nordström spacetimes. *Communications in Mathematical Physics 385*, 1395–1498 (2021)
7. Gajic, D., and Warnick, C. A model problem for quasinormal ringdown on asymptotically flat or extremal black holes. *Journal of Mathematical Physics 61*, 102501 (2020).
8. Angelopoulos, Y., Aretakis, S., and Gajic, D. A non-degenerate scattering theory for the wave equation on extremal Reissner-Nordström spacetimes. *Communications in Mathematical Physics 380*, 323–408 (2020).
9. Angelopoulos, Y., Aretakis, S., and Gajic, D. Nonlinear scalar perturbations of extremal Reissner-Nordström spacetimes. *Annals of PDE 6*, 12 (2020).

10. Angelopoulos, Y., Aretakis, S., and Gajic, D. Late-time asymptotics for the wave equation on extremal Reissner–Nordström backgrounds. *Advances in Mathematics 375* (2020), 107363.
11. Angelopoulos, Y., Aretakis, S., and Gajic, D. Horizon hair of extremal black holes and measurements at null infinity. *Physical Review Letters 121*, 13 (2018), 131102.
12. Angelopoulos, Y., Aretakis, S., and Gajic, D. Asymptotics for scalar perturbations from a neighborhood of the bifurcation sphere. *Classical and Quantum Gravity 35*, 15 (2018), 155007.
13. Angelopoulos, Y., Aretakis, S., and Gajic, D. Logarithmic corrections in the asymptotic expansion for the radiation field along null infinity. *Journal of Hyperbolic Differential Equations 16*, 01 (2019), 1–34.
14. Gajic, D., and Luk, J. The interior of dynamical extremal black holes in spherical symmetry. *Pure and Applied Analysis 1*, 2 (2019), 263–326.
15. Angelopoulos, Y., Aretakis, S., and Gajic, D. Late-time asymptotics for the wave equation on spherically symmetric, stationary backgrounds. *Advances in Mathematics 323* (2018), 529–621.
16. Angelopoulos, Y., Aretakis, S., and Gajic, D. A vector field approach to almost-sharp decay for the wave equation on spherically symmetric, stationary spacetimes. *Annals of PDE 4*, 2 (2018), 15.
17. Angelopoulos, Y., Aretakis, S., and Gajic, D. The trapping effect on degenerate horizons. *Annales Henri Poincaré 18*, 5 (2017), 1593–1633.
18. Gajic, D. Linear waves in the interior of extremal black holes I. *Communications in Mathematical Physics 353* (2017), 717–770.
19. Gajic, D. Linear waves in the interior of extremal black holes II. *Annales Henri Poincaré 18* (2017), 4005–4081.
20. Gajic, D. Linear waves on constant radius limits of cosmological black hole spacetimes. *Advances in Theoretical and Mathematical Physics 22*, 4 (2018), 919–1005.

Preprints

1. Gajic, D., and Warnick, C. Quasinormal modes on Kerr spacetimes. *arXiv:2407.04098*, (2024)
2. Gajic, D. and Van de Moortel, M. Late-time tails for scale-invariant wave equations with a potential and the near-horizon geometry of null infinity. *arXiv:2401.13047*, (2024)
3. Gajic, D. Azimuthal instabilities on extremal black holes. *arXiv:2302.06636*, (2023)
4. Angelopoulos, Y., Aretakis, S., and Gajic, D. Asymptotic blow-up for a class of semi-linear wave equations on extremal Reissner-Nordström. *arXiv:1612.01562*, (2016).

Invited talks

General Relativity Conference: Black holes in General Relativity, from transient behavior to long time dynamics, Fields Institute, University of Toronto, 26–29 May 2026

Extremal Black Holes and the Third Law of Black Hole Thermodynamics, ICERM, Brown University, 5–9 January 2026

Modern Trends in Gravity and Black Holes, University of Crete, 1–7 September 2025

DPG Spring Meeting, University of Göttingen, 31 March–4 April 2025

Singularities in Fluids and General Relativity, National University of Singapore, 6–10 January 2025

Oberwolfach Graduate Seminar: Black Holes and Conformal Infinities of Spacetime, Banach Centre, Będlewo, 13–19 October 2024

XXI International Congress of Mathematical Physics, Strasbourg, 1–6 July 2024

Gravitational physics and its mathematical analysis, SwissMAP, Les Diablerets, 2–7 June 2024

PDE Seminar, University of Cambridge, 13 May 2024

Taking it to the extreme: Symmetries and dynamics of extremal black holes, Princeton University, 6–8 March 2024

Fourth European Physical Society Conference on Gravitation: Black Holes, University of Valencia, 13–15 November 2023

Cretan Waves, University of Crete, 9–13 October 2023

Infinity on a Gridshell, University of Copenhagen, 10–13 July 2023

Inaugural lecture, Faculty Colloquium, University of Leipzig, 27 June 2023

Gravitational Memory Effects: from theory to observation, Queen Mary University, 5–9 June 2023

Oberseminar Analysis, Leipzig University, 20 April 2023

Hyperbolic & Dispersive PDE Seminar, Rutgers University, 23 March 2023

Princeton Gravity Initiative Seminar, Princeton University, 20 March 2023

Ringvorlesung, IMPRS, Leipzig, February 2023

International Conference on Geometric Analysis and Hyperbolic Equations, Guangxi Center for Mathematical Research (online), 10–17 December 2022

Theoretical Physics Seminar, University of Jena, 30 November 2022

PDE seminar, Vanderbilt University, 18 November 2022

Physik-Combo, MPI MiS Leipzig, 5–7 October 2022

Clay Research Conference "Stability and Instability in General Relativity", University of Oxford, 26–30 September 2022

"Topics in general relativity" seminar, WWU Münster, 26 April 2022

IMPRS day, MPI MiS Leipzig, 4 April 2022

GauS Seminar, online, 7 March 2022

Einstein Telescope Seminar, Radboud University, 3 February 2022

Mathematical GR and Hyperbolic PDE seminar, online, 17 December 2021

London PDE Seminar, online, 12 November 2021

New Trends in Geometric PDEs, WWU Münster, 1–5 November 2021

General Relativity and Geometric Analysis Seminar, Columbia University, 29 October 2021

Analysis and PDE seminar, University of Kentucky, 20 October 2021

Oberseminar, MPI MiS Leipzig, 14 September 2021

Oberwolfach workshop "Mathematical Aspects of General Relativity", August–September 2021

Mathematics Online Lunch Seminar, Radboud University, 3 June 2021

Quantum & Gravity Seminar, Radboud University, 9 April 2021

AMS Sectional meeting (online), 21 March 2021,

DAMTP Friday GR seminar, University of Cambridge, 12 February 2021

Analysis Webinar, Loughborough University, 25 November 2020

Analysis Seminar, Caltech, 29 October 2020

Seminar on Quantum Field Theory, Gravitation and Elementary Particles, Leipzig University, 22 June 2020

Analysis Seminar, University of Leeds, 27 May 2020

CMSA General Relativity Seminar, Harvard University, 15 May 2020

Mini-Workshop on Wave Equations, Queen Mary University London, 29 January 2020

International Conference on Geometric Analysis and Hyperbolic Equations, Guangxi Center for Mathematical Research, December 2019

Analysis Seminar, Princeton University, 18 November 2019

General Relativity, Geometry and Analysis: beyond the first 100 years after Einstein, Institut Mittag-Leffler, 5 November 2019

PDE CDT Lunchtime seminar, University of Oxford, 17 October 2019

Analysis and PDE Seminar, University of California, Berkeley, 29 April 2019

Analysis & PDE Seminar, National University of Singapore, 28 March 2019

Junior Analysis Seminar, Imperial College London, 16 November 2018

Conference on Mathematical General Relativity, IHP, Paris, 28 May– 1 June 2018

DAMTP Friday GR seminar, University of Cambridge, 23 February 2018

Analysis and PDE seminar, Stanford University, 4 December 2017

Analysis Seminar, University of Edinburgh, 20 November 2017

Seminar on Mathematical General Relativity, UPMC, Paris, 6 June 2017

Analysis Seminar, Kings College London, 18 May 2017

Analysis Seminar, Princeton University, 24 April 2017

Warwick EPSRC Symposium: Geometric PDEs, University of Warwick, 12–16 December 2016

Pure analysis and PDE seminar, Imperial College London, 21 October 2016

Mathematical physics seminar, University of Grenoble, 18 May 2016

PDE seminar, CAMGSD, University of Lisbon, 4 May 2016

PDE CDT Lunchtime Seminar, University of Oxford, 10 March 2016

DAMTP Friday GR seminar, University of Cambridge, 26 February 2016

Junior Warwick–Imperial–Cambridge Geometric Analysis seminar, Queen Mary University of London, 6 January 2016

Focus Program on 100 Years of General Relativity, Fields Institute, University of Toronto, 9 June 2015